

## ESENGULY, Turkmenistan

WMO: 387500

Lat: 37.467N

Lon: 53.967E

Elev: -72

StdP: 14.73

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 27.4       | 30.5 | 20.8                          | 15.6 | 34.6 | 24.6 | 18.6 | 37.2 | 18.6                  | 48.0 | 16.6 | 48.4 | 4.8                   | 20   | 0.431 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.4                   | 93.5            | 76.1 | 91.5 | 76.8 | 89.9 | 76.5 | 82.3                | 88.4 | 80.9 | 87.7 | 79.7 | 86.9 | 8.3                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.7                            | 158.7 | 86.6 | 78.9 | 149.6 | 86.2 | 77.5 | 142.7 | 85.5 | 46.1          | 88.5 | 44.5 | 87.7 | 43.3 | 87.1 | 90.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 20.1              | 17.1 | 14.9 | 22.5                       | 102.1 | 5.2                | 4.8 | 18.8                                               | 105.5 | 15.8       | 108.4 | 12.9       | 111.1 | 9.1        | 114.6 |
|                   |      |      | 21.2                       | 84.1  | 4.8                | 2.5 | 17.8                                               | 85.8  | 15.0       | 87.3  | 12.3       | 88.7  | 8.9        | 90.5  |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 63.7  | 44.1  | 45.5  | 52.1  | 60.4  | 70.6  | 78.2  | 82.9  | 83.9  | 77.2  | 66.6  | 54.6  | 46.8 | (6)  |
|      |                                                                   | DBStd       | 15.27 | 5.74  | 6.19  | 6.01  | 6.14  | 5.30  | 4.09  | 3.23  | 2.99  | 4.53  | 5.99  | 6.78  | 5.70 | (7)  |
|      |                                                                   | HDD50       | 545   | 197   | 144   | 40    | 2     | 0     | 0     | 0     | 0     | 0     | 29    | 133   |      | (8)  |
|      |                                                                   | HDD65       | 2710  | 649   | 545   | 403   | 165   | 11    | 0     | 0     | 0     | 54    | 317   | 566   |      | (9)  |
|      |                                                                   | CDD50       | 5535  | 12    | 19    | 106   | 314   | 639   | 845   | 1019  | 1052  | 817   | 514   | 166   | 32   | (10) |
|      |                                                                   | CDD65       | 2226  | 0     | 0     | 4     | 27    | 185   | 395   | 554   | 587   | 368   | 102   | 4     | 0    | (11) |
|      |                                                                   | CDH74       | 23662 | 0     | 5     | 80    | 283   | 1186  | 3613  | 6609  | 7576  | 3614  | 653   | 39    | 4    | (12) |
|      |                                                                   | CDH80       | 8962  | 0     | 0     | 25    | 100   | 365   | 1098  | 2620  | 3475  | 1134  | 138   | 7     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 6.4   | 5.0   | 5.9   | 6.8   | 7.4   | 7.6   | 8.0   | 8.0   | 7.0   | 6.1   | 5.3   | 4.8   | 4.7  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 8.1   | 0.9   | 0.8   | 1.1   | 0.7   | 0.5   | 0.2   | 0.3   | 0.3   | 0.5   | 0.7   | 1.0   | 1.0  | (15) |
| (16) |                                                                   | PrecMax     | 12.6  | 2.9   | 2.3   | 4.1   | 2.1   | 2.0   | 1.3   | 1.5   | 1.8   | 2.2   | 3.6   | 2.7   | 2.3  | (16) |
| (17) |                                                                   | PrecMin     | 3.3   | 0.1   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  | (17) |
| (18) |                                                                   | PrecStd     | 2.3   | 0.7   | 0.5   | 0.7   | 0.6   | 0.6   | 0.4   | 0.4   | 0.4   | 0.6   | 0.9   | 0.7   | 0.6  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 67.7  | 72.2  | 82.8  | 89.8  | 95.0  | 95.3  | 95.8  | 96.4  | 92.9  | 90.1  | 78.9  | 71.6  |      | (19) |
| (20) |                                                                   | 0.4% MCWB   | 53.0  | 54.6  | 59.2  | 63.6  | 67.6  | 71.2  | 77.7  | 77.4  | 74.0  | 65.9  | 62.3  | 56.4  |      | (20) |
| (21) |                                                                   | 2% DB       | 60.6  | 63.9  | 73.9  | 82.2  | 87.7  | 90.0  | 92.2  | 93.5  | 89.6  | 83.4  | 72.4  | 63.5  |      | (21) |
| (22) |                                                                   | 2% MCWB     | 50.6  | 52.7  | 57.0  | 62.4  | 67.7  | 74.4  | 78.2  | 78.2  | 74.4  | 66.8  | 61.2  | 53.6  |      | (22) |
| (23) |                                                                   | 5% DB       | 55.9  | 58.6  | 67.6  | 76.0  | 83.5  | 87.7  | 90.4  | 91.7  | 87.6  | 80.1  | 68.6  | 59.2  |      | (23) |
| (24) |                                                                   | 5% MCWB     | 49.3  | 49.8  | 55.7  | 61.0  | 67.5  | 74.3  | 77.9  | 78.5  | 73.9  | 67.0  | 59.3  | 52.4  |      | (24) |
| (25) |                                                                   | 10% DB      | 53.0  | 54.8  | 62.0  | 71.2  | 80.0  | 85.6  | 88.7  | 90.2  | 85.8  | 77.4  | 65.5  | 56.2  |      | (25) |
| (26) |                                                                   | 10% MCWB    | 48.0  | 48.5  | 53.1  | 60.1  | 67.2  | 73.9  | 77.7  | 78.0  | 73.4  | 66.1  | 57.9  | 50.9  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 55.3  | 57.0  | 62.4  | 67.6  | 74.9  | 82.5  | 84.4  | 85.2  | 80.5  | 73.1  | 66.1  | 58.3  |      | (27) |
| (28) |                                                                   | 0.4% MCDB   | 63.3  | 66.9  | 76.3  | 80.7  | 82.7  | 86.4  | 88.6  | 89.9  | 87.6  | 79.2  | 73.9  | 65.7  |      | (28) |
| (29) |                                                                   | 2% WB       | 52.3  | 53.6  | 58.7  | 65.1  | 72.3  | 79.0  | 82.3  | 82.4  | 78.6  | 70.9  | 63.0  | 55.3  |      | (29) |
| (30) |                                                                   | 2% MCDB     | 58.2  | 61.5  | 70.7  | 76.0  | 80.3  | 84.9  | 87.3  | 89.7  | 86.2  | 78.2  | 69.4  | 61.1  |      | (30) |
| (31) |                                                                   | 5% WB       | 50.3  | 51.1  | 56.1  | 63.0  | 70.5  | 77.5  | 80.7  | 81.1  | 77.1  | 69.6  | 60.9  | 53.3  |      | (31) |
| (32) |                                                                   | 5% MCDB     | 55.1  | 56.6  | 65.3  | 72.6  | 78.6  | 84.1  | 87.2  | 88.4  | 84.6  | 76.9  | 66.6  | 58.6  |      | (32) |
| (33) |                                                                   | 10% WB      | 48.4  | 49.4  | 53.9  | 61.4  | 68.9  | 76.1  | 79.4  | 80.1  | 75.8  | 68.0  | 58.8  | 51.2  |      | (33) |
| (34) |                                                                   | 10% MCDB    | 52.4  | 53.9  | 60.9  | 69.5  | 77.1  | 83.0  | 86.5  | 87.5  | 83.4  | 75.3  | 64.2  | 55.9  |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 16.7  | 16.4  | 16.1  | 15.3  | 14.2  | 12.1  | 9.5   | 11.4  | 14.8  | 18.3  | 17.7  | 16.7  |      | (35) |
| (36) |                                                                   | 5% DB MCDBR | 22.7  | 25.8  | 29.9  | 27.7  | 22.4  | 15.7  | 12.2  | 14.7  | 17.4  | 22.2  | 22.9  | 22.6  |      | (36) |
| (37) |                                                                   | 5% WB MCWBR | 14.5  | 15.4  | 14.6  | 11.9  | 8.7   | 6.6   | 5.9   | 7.4   | 9.3   | 11.3  | 13.6  | 15.0  |      | (37) |
| (38) |                                                                   | 5% WB MCDBR | 20.1  | 21.0  | 24.4  | 21.2  | 16.0  | 11.7  | 10.0  | 12.3  | 13.5  | 17.3  | 18.2  | 19.7  |      | (38) |
| (39) |                                                                   | 5% WB MCWBR | 13.9  | 13.8  | 13.2  | 10.3  | 7.9   | 5.8   | 5.6   | 7.0   | 8.5   | 10.5  | 12.3  | 13.9  |      | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.405 | 0.452 | 0.509 | 0.579 | 0.580 | 0.591 | 0.634 | 0.590 | 0.566 | 0.529 | 0.454 | 0.411 |      | (40) |
| (41) |                                                                   | taud        | 2.121 | 1.998 | 1.878 | 1.769 | 1.808 | 1.813 | 1.748 | 1.826 | 1.849 | 1.899 | 2.038 | 2.113 |      | (41) |
| (42) |                                                                   | Ebn at Noon | 236   | 239   | 238   | 229   | 232   | 229   | 218   | 224   | 220   | 212   | 215   | 222   |      | (42) |
| (43) |                                                                   | Edh at Noon | 37    | 48    | 59    | 69    | 68    | 68    | 71    | 65    | 60    | 52    | 40    | 35    |      | (43) |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 756   | 985   | 1310  | 1633  | 2086  | 2345  | 2215  | 2079  | 1696  | 1224  | 847   | 668   |      | (44) |
| (45) |                                                                   | RadStd      | 52    | 96    | 81    | 176   | 99    | 93    | 121   | 133   | 68    | 62    | 46    | 48    |      | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |      |
| (46) | Station Only           | +0.94 | N/A     | N/A    | +1.75   | +1.10 | +1.26 | N/A         | N/A   | N/A   | N/A   | (46) |
| (47) | Regional (5 neighbors) | +0.56 | N/A     | N/A    | +1.03   | +0.56 | +0.58 | N/A         | N/A   | N/A   | N/A   | (47) |

Nomenclature: See separate page